

## Political Correctness, Social Image, and Information Transmission<sup>†</sup>

By LUCA BRAGHIERI\*

*A prominent argument in the political correctness debate is that people feel pressure to publicly espouse sociopolitical views they do not privately hold, and that such misrepresentations might render public discourse less vibrant and informative. This paper formalizes the argument in terms of social image and evaluates it experimentally in the context of college campuses. The results show that (i) social image concerns drive a wedge between the sensitive sociopolitical attitudes that college students report in private and in public; (ii) public utterances are indeed less informative than private utterances; and (iii) information loss is exacerbated by (partial) audience naïveté. (JEL D72, D83, D91, I23, Z13)*

Political correctness has been at the center of a heated debate in the United States and abroad for more than 30 years (Lea 2009; Roth 2019). The terms of the debate are in constant evolution, but a prominent and recurrent argument is that people may feel social pressure to publicly espouse views on a set of sensitive political topics that they may not privately hold, and that such misrepresentations may render public discourse less vibrant and informative (Ackerman et al. 2020; Lea 2009; Loury 1994; Morris 2001; Schwarz 2020; Zimmer 2016). Despite the wealth of articles and books making some version of the argument above, there is little direct evidence assessing its merits.<sup>1</sup>

\*Braghieri: Bocconi University (email: [luca.braghieri@unibocconi.it](mailto:luca.braghieri@unibocconi.it)). Stefano DellaVigna was the coeditor for this article. I thank Matthew Gentzkow and B. Douglas Bernheim for invaluable advising. I thank Leonardo Bursztyn, Davide Cantoni, Sarah Eichmeyer, Uri Gneezy, Muriel Niederle, Frank Schilbach, Klaus Schmidt, David Yanagizawa-Drott, David Yang, and seminar participants at Berkeley Haas, Chicago Booth, Columbia Business School, Harvard, Harvard Business School, and MIT for helpful comments. I thank Aman Khinvasara and Dhruv Jatkar for excellent research assistance. I gratefully acknowledge financial support from the George P. Shultz Dissertation Support Fund Fellowship, the B.F. Haley and E.S. Shaw Fellowship for Economics, the Stanford Graduate Research Opportunity Fund, the Stanford Center for American Democracy, the Stanford Institute for Research in the Social Sciences (IRiSS), the Ludwig Maximilian University of Munich (CRC TRR 190), and Bocconi University. The study was approved by the Institutional Review Boards at Bocconi University (Protocol SA000600). The experiments (Braghieri 2023a,b) were registered in the American Economic Association Registry for randomized control trials under trial number AEARCTR-0011440 (<https://doi.org/10.1257/rct.11440-1.1>) and AEARCTR-0011981 (<https://doi.org/10.1257/rct.11981-1.0>). Survey instruments can be found in the online Appendix Replication files and survey instruments are available on ICPSR (Braghieri 2024). Disclosures: I have no relevant or material disclosures.

<sup>†</sup>Go to <https://doi.org/10.1257/aer.20210039> to visit the article page for additional materials and author disclosure statement(s).

<sup>1</sup>A version of the argument was recently made by Nobel Prize-winning economist James Heckman (Schwarz 2020); by the former president of the University of Chicago, Robert Zimmer (Zimmer 2016); and by many prominent academics and writers (including Noam Chomsky, Francis Fukuyama, Jonathan Haidt, Deirdre McCloskey, and Steven Pinker) who signed an open letter published in *Harper's Magazine* titled “A Letter on Justice and Open Debate” (Ackerman et al. 2020).

I argue that, in order to generate empirical evidence in favor or against the argument, one needs to address three key questions. First, when it comes to topics related to political correctness, do social image concerns drive a wedge between the attitudes that speakers portray in public and the ones they hold in private? Second, does the existence of such wedge reduce the amount of information that public statements convey even to a hypothetical sophisticated audience who have a clear and accurate understanding of the ways in which social image concerns distort publicly portrayed attitudes? Third, is information transmission further impaired by the fact that the actual audience in the environment might neglect or misunderstand the ways in which social image concerns distort public attitudes?

The following example illustrates the importance of addressing all three of the questions above when thinking about impairments to information transmission in the context of the political correctness debate. Consider a faculty council in which faculty members discuss in an open forum the value of diversity, equity, and inclusion (DEI) initiatives. Imagine three scenarios. First, suppose speakers speak truthfully—in the sense that their public statements reflect their private attitudes—and listeners understand that. Then, all information about the speakers' private attitudes is trivially transmitted to the audience at the faculty council. Second, suppose that, in order to protect their social image, all speakers voice the same opinion in public in a manner similar to how players behave in the pooling equilibrium of a standard signaling game. Then, even a hypothetical sophisticated listener who has a clear and accurate understanding of the ways in which social image concerns distort public attitudes would be unable to extract any information from public statements. Third, suppose speakers speak truthfully, but listeners incorrectly believe that speakers publicly portray views that are more socially acceptable than the ones they actually hold. Then, public statements would be perfectly informative of the speakers' privately held attitudes from the perspective of the hypothetical sophisticated audience, but they would not be equally informative from the perspective of the actual audience. As a result of their incorrect understanding of the mapping between speakers' private opinions and public statements, the actual audience will draw incorrect inferences about the distribution of private opinions and might leave the faculty council believing that there is less support for DEI initiatives than there actually is. Such misperceptions might in turn affect downstream choices, such as hiring decisions and tenure cases.

The following simple model of communication underlies the experimental investigation. The model involves a speaker and an audience. The speaker has a private opinion about a topic (for instance, the value of DEI initiatives) and makes a public statement. The audience does not know the speaker's private opinion about the topic but hears the speaker's public statement and uses it to draw inferences about the speaker's private opinion.

Even in the simple and stylized model of communication above, two fundamental processes are at play: a process of encoding, which describes the way in which the speaker maps her private opinions into public statements, and a process of decoding, which describes, for each possible public statement, the receiver's inferences about the speaker's private opinion. As I discuss in detail in a companion paper (Braghieri 2023), the amount of information that is transmitted to the audience is

a function of both processes. Specifically, in order for the audience to be able to extract information from the speakers' public statements, some information has to be encoded in those statements in the first place. In the context of the experiment, the amount of information encoded in the speakers' public statements can be thought of as the amount of information conveyed to a hypothetical sophisticated audience who understands the ways in which social image concerns distort the attitudes that students portray in public. Unlike the hypothetical sophisticated audience, however, the actual audience in the experiment, namely other students, might misunderstand the ways in which social image affects their peers' public attitudes and might decode public statements incorrectly. In that case, the amount of information transmitted to the actual audience would be strictly lower than the amount of information encoded.<sup>2</sup>

This simple model of communication provides a rationale for structuring the paper around the three questions posed at the beginning of the introduction. Using the terminology introduced in the previous paragraph, the three questions I ask in this paper can be rephrased as: (i) Do social image concerns around sensitive topics related to the political correctness debate lead students to publicly state opinions that they do not privately hold? (ii) How much information is lost, if any, in the process of encoding? (iii) Is there additional information lost in the process of decoding?

In order to answer the first question, I ran three waves of an experiment—which I refer to henceforth as the Encoding experiment—at two universities in the University of California system: the University of California Santa Barbara (UCSB) and the University of California San Diego (UCSD). In all three waves of the experiment, I manipulated students' social image concerns by varying the perceived observability of their answers. In one of the treatments (Private treatment), participants were assured that their answers to the survey would remain completely anonymous; in the other treatment (Public treatment), participants were not given such assurance of anonymity and were in fact given hints suggesting that their individual-level answers might be shared with other students from their college, together with their names. The survey asked all subjects to report, on a scale from 0 to 10, the extent to which they agreed with a set of sensitive statements related to recurrent topics in the political correctness debate. The statements covered issues such as the removal of Confederate statues, the use of preferred gender pronouns and trigger warnings in classrooms, illegal immigration, culturally insensitive Halloween costumes, and more.

The first set of experimental results shows that social image concerns do indeed drive a wedge between the sensitive political attitudes that students at UCSB and UCSD report in private and in public. Specifically, pooling the results of the three waves of the experiment, I find that the average level of agreement with the sensitive statements of subjects in the Public treatment is 0.13 standard deviations closer to the end of the spectrum that is generally perceived to be more socially acceptable on campus than that of subjects in the Private treatment.

<sup>2</sup>See Braghieri (2023) for details, especially Section III.

In order to answer the second question, namely, how much information is lost in the encoding process, I rely on a set of theoretical notions of informativeness. In particular, as suggested by a signaling model developed in online Appendix A, one can measure information loss in the encoding process by studying the relative extent to which the answers to the sensitive questions given by students assigned to the Private and the Public treatment are predictive of the students' demographic characteristics and incentivized behaviors. The results show that the predictions made using the answers of students assigned to the Private treatment are significantly more accurate than the predictions made using the answers of students assigned to the Public treatment. In particular, employing a canonical measure of information transmission (*mutual information*), I show that the degree to which the students' statements are informative of their demographic characteristics and incentivized behaviors is 30 to 50 percent lower in the Public treatment than in the Private treatment. Therefore, the answers of participants in the Public treatment are less informative than the answers of participants in the Private treatment even from the perspective of a sophisticated audience that perfectly understands the ways in which social image affects students' public statements.

In order to answer the third question, namely, whether there is additional information loss in the decoding process, I recruited a separate group of students and studied whether they take the effects of social image into account when interpreting their peers' public statements. At its core, the experiment—referred to henceforth as the Decoding experiment—involved asking students to forecast the results of the Encoding experiment and incentivizing them for accuracy.

The first part of the Decoding experiment aimed to test whether students take the effects of social image into account when social image is not made salient. Specifically, for their first set of forecasts, participants in the Decoding experiment were made aware of the existence of only one of the treatments in the Encoding experiment; therefore, social image was arguably not very salient. Conversely, for their second set of forecasts, they were informed that the Encoding experiment had two treatments, and they were shown the instructions for both. Revealing the existence of the two treatments of the Encoding experiment was supposed to draw the forecasters' attention to social image, thus increasing its salience. The first main result of the Decoding experiment is that students take into account the role of social image in shaping their peers' public statements only when social image is made salient.

For the rest of the Decoding experiment, participants were aware of the fact that the Encoding experiment had two treatments, thus allowing me to study how students interpret their peers' public statements when social image is salient. Specifically, for each of the statements from the Encoding experiment, participants were shown the distribution of answers of subjects assigned to the Public treatment and were asked to forecast the average answer of subjects assigned to the Private treatment. The second main result of the Decoding experiment is that participants exhibit a qualitatively accurate understanding of the ways in which social image distorts their peers' public answers to the sensitive questions, but they overestimate the magnitude of such distortions. The last result of the Decoding experiment is that students fail to recognize that the effects of social image are highly heterogeneous by political ideology, thus misunderstanding a crucial dimension of how social image concerns operate on their college campus.

This project relates to the empirical literature in economics on the effects of social image.<sup>3</sup> The three papers that are most closely related to this project are Bursztyn, Fujiwara, and Pallais (2017); Bursztyn, González, and Yanagizawa-Drott (2020); and Bursztyn, Egorov, and Fiorin (2020). Bursztyn, Fujiwara, and Pallais (2017) shows that, in a high-stakes job placement survey, female MBA students indicated a preference for lower salaries and a lesser inclination to engage in extensive travel and overtime when their answers were observable by their classmates than when they were private. Bursztyn, González, and Yanagizawa-Drott (2020) shows that young men in Saudi Arabia underestimated the level of support for female labor force participation (FLFP) among their peers, and that an information intervention that randomly disclosed the actual popularity of FLFP increased married men's willingness to let their wives join the labor force. Bursztyn, Egorov, and Fiorin (2020) shows that Donald Trump's rise in popularity and eventual victory in the 2016 presidential election increased the proportion of people willing to publicly display anti-immigration attitudes and suggests the results are due to the election precipitating common knowledge about the prevalence of anti-immigration sentiments in the population.

This paper can be thought of as studying the layer in between Bursztyn, Fujiwara, and Pallais (2017) and the other two papers. Specifically, Bursztyn, Fujiwara, and Pallais (2017) provides an example of individuals misrepresenting their preferences when in public in order to shape the audience's inferences about their private characteristics. The latter two papers study the behavioral implications of people holding imprecise and incorrect beliefs in politically sensitive contexts and suddenly updating their perceptions of the attitudes held by their peers. My paper links the two layers by studying the process of belief formation when public statements are affected by social image. Specifically, I investigate (i) the degree to which misrepresenting one's preferences in public as in Bursztyn, Fujiwara, and Pallais (2017) reduces the informativeness of public statements, thus leading even a sophisticated audience to hold imprecise beliefs, and (ii) the inferences that the actual audience in the environment draws from public statements, which might produce systematic misperceptions akin to the ones documented in Bursztyn, González, and Yanagizawa-Drott (2020) and Bursztyn, Egorov, and Fiorin (2020).

Finally, this paper relates to the large literature in the social sciences about the phenomenon of social desirability bias in survey responses (Edwards 1957; Folsom et al. 1973; Greenberg et al. 1971; Krumpal 2013; Raghavarao and Federer 1979; Sniderman and Piazza 1993; Sudman and Bradburn 1974; Tourangeau, Rips, and Rasinski 2000; Warner 1965, 1971). Sudman and Bradburn (1974, pp. 9–10) summarize the phenomenon as follows: "Some questions call for the respondent to provide information on topics that have highly desirable answers [ . . . ] If the respondent has a socially undesirable attitude or if he has engaged in socially undesirable behavior, he may face a conflict between a desire to tell the truth and a desire to appear to the interviewer to be in the socially desirable category." This paper contributes

<sup>3</sup> See, for instance, Alpizar, Carlsson, and Johansson-Stenman (2008); Banerjee et al. (2024); Bursztyn et al. (2018); Butera et al. (2022); Chandrasekhar, Golub, and Yang (2019); DellaVigna, List, and Malmendier (2012); DellaVigna et al. (2017); Gerber, Green, and Larimer (2008); Karing (forthcoming); Kosfeld and Neckermann (2011); Perez-Truglia and Cruces (2017); and Perez-Truglia and Troiano (2018).



to the literature on social desirability bias in surveys by showing (i) that Sudman and Bradburn's (1974) intuition can be neatly captured in the formal game theoretic model of signaling with lying costs and social image concerns developed in online Appendix A, (ii) that such model can sharpen our understanding of the relationship between social desirability bias and information loss, and (iii) that incorrect decodings by readers and analysts can further impair information extraction from survey responses tainted by social desirability bias.

The remainder of the paper is organized as follows. Section I introduces the motivating framework. Section II presents the design of the Encoding experiment and shows that social image concerns distort the attitudes that students report in public compared to the ones they report in private. Section III discusses the implications of such distortions for information loss in the encoding process. Section IV presents the design of the Decoding experiment and studies whether there is additional information loss in the decoding process. Section V concludes.

### **I. Motivating Framework**

The theoretical framework underlying the experiment is developed formally in Braghieri (2023). This section presents the main intuition. Consider a speaker and an audience. The speaker has a private opinion about a topic and makes a public statement. The audience does not know the speaker's private opinion about the topic but hears the speaker's public statement and uses it to draw inferences about the speaker's private opinion. Even this simple model of communication features two fundamental and distinct processes: a process of encoding, which describes the stochastic mapping between the speaker's private opinions and public statements, and a process of decoding, which describes the audience's inferences about the speaker's private opinion given her public statements. As discussed in detail in Braghieri (2023), information transmission is a function of both processes.<sup>4</sup>

The stylized model above provides a formal justification for the way in which the paper is organized. Specifically, information loss in the encoding process is akin to information loss from the perspective of a sophisticated audience who understand the ways in which social image concerns distort the students' public statements (Section III). Conversely, information loss in the decoding process relates to the potentially incorrect inferences of the actual audience in the environment (Section IV).

When studying information loss in the encoding process, it is possible to embed a signaling model with lying costs along the lines of Kartik (2009) in the simple communication framework described above. Such model is developed in online Appendix A. At its core, the model features an agent who trades off the cost of misrepresenting her private information against the benefit of garnering social esteem by means of the misrepresentation. The equilibrium of the model determines the extent to which statements are informative of the agent's private attitudes from the perspective of a sophisticated audience. The equilibrium analysis helps link a canonical and appealing notion of informativeness, namely the Blackwell partial order, to the empirical measures deployed in Section III.

<sup>4</sup>See, in particular, Section III of Braghieri (2023).

## II. Effects of Social Image on the Attitudes Reported by Students

The effects of social image on the attitudes reported by college students are studied in the context of the Encoding experiment. The design of the Encoding experiment had two core objectives. First, to allow me to study whether, for a set of sensitive political topics, social image concerns distort the attitudes that students portray in public. Second, to allow me to study whether the distortions induced by social image lead to information loss in the encoding process.

In order to meet these objectives, I elicited two sets of variables from students in the Encoding experiment: (i) the students' reported levels of agreement with various statements and (ii) some demographic characteristics and an incentivized behavioral measure. The students' reported levels of agreement with the statements allow me to study whether, for the sensitive but not for the placebo statements, social image concerns drive a wedge between the attitudes that students portray in private and in public. The demographic characteristics and the incentivized behavioral measure allow me to study the degree of informativeness of the students' answers in the Private and Public treatments from the perspective of a sophisticated audience. Specifically, one of the demographic characteristics (political affiliation) and the behavioral measure (a donation to a charity that promotes transgender rights) are expected to correlate with the students' private attitudes toward some of the sensitive statements in the experiment. To test whether the distortions caused by social image concerns lead to information loss even for a sophisticated audience, I study whether the answers of students in the Private treatment are better predictors of the students' political affiliations and donation rates than the answers of students in the Public treatment.<sup>5</sup>

### A. Encoding Experiment

*Logistics.*—I ran three waves of the Encoding experiment: one in 2019, before the onset of the COVID-19 pandemic, one in 2022, close to the peak of the COVID-19 pandemic, and one in 2023, well after the peak of the COVID-19 pandemic. The three waves of the Encoding experiment were run at two universities in the University of California system: the University of California Santa Barbara and the University of California San Diego. According to Niche, a company that gathers information about colleges to generate college rankings, both UCSB and UCSD fall in the top quintile in an ordered list of colleges from most liberal to most conservative (Niche 2020).<sup>6</sup> Among colleges that are at least as selective as UCSB and UCSD in terms of admission rates, UCSB and UCSD do not seem to be outliers in terms of the political leaning of their students. Online Appendix Table A1 shows a list of selected universities that, together with UCSB and UCSD, are found in the top quintile of the liberal–conservative distribution according to the Niche ranking.

<sup>5</sup>The theoretical model in online Appendix A explains in detail the rationale behind this approach.

<sup>6</sup>The ranking is calculated by surveying a sample of students from each college and asking them both about their personal political leanings and about their beliefs about the political leanings of the other students at the college (Niche 2020). In the ranking, the first percentile corresponds to the most liberal colleges.

*Experimental Design.*—In this section, I describe the design of the last wave of the Encoding experiment, namely the one run at UCSD in 2023. The design of the other two waves of the Encoding experiment is identical, other than for the features highlighted in online Appendix Table A2. The full instructions of the third wave of the Encoding experiment are available in online Appendix H.

I recruited subjects through the experimental laboratory at the University of California San Diego to complete a short online survey. The recruitment email did not mention the topic of the study. Before being directed to the consent form, subjects were asked to complete a brief prescreen questionnaire that contained a few demographic questions and a short news-knowledge quiz. The aim of the prescreen questionnaire was to screen out ineligible and low-quality survey takers. A subject was considered eligible to participate in the study if she reported being an undergraduate student at UCSD; the subject was considered a high-quality survey taker if she spent at least 30 seconds on the news-knowledge quiz.

After completing the prescreen survey and consenting to participate in the experiment, subjects were informed that they would be entered into a lottery. The lottery would randomly select a participant to receive \$100 and play a dictator game with Athlete Ally, a national not-for-profit organization that, among other activities, advocates for the inclusion of transgender women in women's sports. All subjects were asked how much of the \$100 they would be willing to anonymously share with Athlete Ally if they were randomly selected and were informed that, if they were indeed selected, their decision would be automatically implemented.

Next, subjects were randomized in equal proportions into one of two treatments. In both treatments, subjects were informed that the results of the subsequent section of the survey would be shared with approximately 200 UCSD students scheduled to participate in the next phase of the study.<sup>7</sup> Importantly, in one of the treatments (Private treatment), subjects were informed that only aggregate-level answers would be shared with participants in the next phase of the study and were given a guarantee of anonymity. Specifically, subjects assigned to the Private treatment were shown a screen that read,

**IMPORTANT:** We will share **aggregate-level answers** to the questionnaire on the next screen with approximately 200 UCSD students who are scheduled to participate in the next phase of the study but no-one, not even the research team, will match your individual answers to any information that may identify you. **Your answers to this survey are thus completely anonymous.**

In contrast, subjects assigned to the other treatment (Public treatment) were told that their individual-level answers would be shared with participants in the next phase of the study and were not given a guarantee of anonymity. In fact, as in Bursztyn, Egorov, and Fiorin (2020), subjects in the Public treatment were shown instructions designed to lead them to doubt that their answers would be kept confidential. Specifically, subjects assigned to the Public treatment were shown a screen that read,

<sup>7</sup>The answers of participants in the Encoding experiment were indeed shared with approximately 200 UCSD students who participated in the Decoding experiment.



**IMPORTANT:** We will share **your individual answers** to the questionnaire on the next screen, as well as the individual-level answers of the other participants in this phase of the study, with approximately 200 UCSD students who are scheduled to participate in the next phase of the study. **There is no need to provide your first and last name here; your information is already in the UCSD laboratory recruitment system.**

[Page break]

According to the UCSD laboratory recruitment system, your first and last name are:

**First Name:** [Participant's First Name]

**Last Name:** [Participant's Last Name]

In practice, even in the treatment I refer to as the Public treatment, the subjects' answers were fully anonymized before being shared with participants in the subsequent phase of the study, as required by the Institutional Review Board at Bocconi University. Importantly, however, the subjects' answers were not reported in aggregate form; instead, each subject's individual-level answers, together with the subject's (anonymous) participant ID, were shared with participants in the subsequent phase of the study.<sup>8</sup>

The experimental manipulation described above aimed to vary the extent to which subjects were concerned about social image when answering a subsequent set of questions. Subjects in the Private treatment were not expected to be very concerned with seeking social approbation when answering the questions because of the guaranteed anonymity of their responses. Conversely, doubts about the confidentiality of their responses were expected to induce subjects in the Public treatment to skew their answers in the direction that they thought would induce the UCSD students scheduled to participate in the next phase of the study to perceive them more favorably.<sup>9</sup>

After being assigned to the Private or the Public treatment and reading the corresponding instructions, participants were asked to report, on sliders from 0 to 10, where 0 stood for "strongly disagree" and 10 stood for "strongly agree," the extent to which they agreed or disagreed with 15 statements. The order of the statements was randomized at the participant level. Of the 15 statements, 10 were preregistered as sensitive (and thus likely affected by social image), and 5 were preregistered as placebos (and thus likely not affected by social image). The sensitive and placebo statements are presented in Table 1.

<sup>8</sup> Notice the statements in both conditions are factually true: the UCSD laboratory recruitment system does contain the first and last name of each student that participated in the experiment, and the answers of participants in the Private treatment were in fact anonymous because the answers of those students were identified only by an arbitrary participant number. The manipulation is similar to the one in Bursztyn, Egorov, and Fiorin (2020).

<sup>9</sup> Notice it is not necessary for subjects in the Private treatment to believe with probability one that their answers will be kept anonymous. For the experimental manipulation to have an effect, it is only necessary that subjects in the Public treatment be sufficiently more concerned about social image when answering the questions than subjects in the Private treatment. Furthermore, notice that the empirical analysis, including the analysis of information loss in the encoding process, does not require that students assigned to the Private treatment truthfully report their private level of agreement with each of the statements.

TABLE 1—LIST OF SENSITIVE AND PLACEBO STATEMENTS

| <i>Sensitive statements</i> |                                                                                                                                                              |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                           | All statues and memorials of Confederate leaders should be removed.                                                                                          |
| 2                           | Defunding the police is a bad idea because it will inevitably lead to increased crime rates. <sup>a,b</sup>                                                  |
| 3                           | The UCSD administration should require professors to address students according to the students' preferred gender pronouns.                                  |
| 4                           | Transgender women (i.e., individuals who were male at birth and transitioned later in life) should be allowed to participate in women's sports. <sup>b</sup> |
| 5                           | The UCSD administration should require professors to use trigger warnings in their classes.                                                                  |
| 6                           | Sexual harassment training should be mandatory for everybody who works or studies at UCSD.                                                                   |
| 7                           | People who immigrated to the US illegally, when caught, should be deported and sent back to their countries of origin. <sup>a</sup>                          |
| 8                           | The US government should provide reparations for slavery.                                                                                                    |
| 9                           | Racial microaggressions are an important problem at UCSD.                                                                                                    |
| 10                          | The UCSD administration should allow students to wear blackface for Halloween. <sup>a</sup>                                                                  |
| <i>Placebo statements</i>   |                                                                                                                                                              |
| 11                          | Parents should limit the amount of time their kids spend on their smartphones.                                                                               |
| 12                          | The United States should increase tariffs on foreign imports.                                                                                                |
| 13                          | School uniforms help reduce clothing-related peer pressure. <sup>a</sup>                                                                                     |
| 14                          | The one-cent coin (i.e., the penny) should be removed from circulation. <sup>a</sup>                                                                         |
| 15                          | The members states of the European Union should cede more powers to the E.U.                                                                                 |

*Notes:* This table presents the 15 statements shown to participants in wave three of the Encoding experiment. Some of the statements were preceded by a brief paragraph providing additional information. For instance, one such paragraph explained that “trigger warnings” are warnings that some work contains writing, images, or concepts that may be distressing to some people. See the survey instrument in online Appendix H for details.

<sup>a</sup>In the context of an exploratory survey run in support of wave three of the Encoding experiment, the fraction of respondents who answered that agreeing with the statement is more socially acceptable on campus than disagreeing with it is smaller than the fraction of participants who answered the opposite.

<sup>b</sup>Statements were included in wave three of the Encoding experiment but not in waves one and two.

*Outcome Variables.*—The outcome variables in the Encoding experiment are the *donation rates* to Athlete Ally and the *levels of agreement* with each of the sensitive and placebo statements.

The levels of agreement with each of the sensitive and placebo statements are reoriented in such a way that more positive values always indicate views that are generally perceived to be more socially acceptable on campus.<sup>10</sup> I summarize the participants' attitudes toward the sensitive (placebo) statements in an equally weighted index by taking a simple average of the participants' reoriented levels of agreement with each of the sensitive (placebo) statements.

## B. Descriptive Statistics and Empirical Strategy

*Descriptive Statistics.*—Online Appendix Table A3 lists the number of subjects who participated in each wave of the Encoding experiment. Online Appendix Table A4 pools data across the three waves of the Encoding experiment and displays, for each of the sensitive statements included in all three waves, summary statistics about the levels of agreement of participants in the Private treatment. Online Appendix Table A5 shows that, for the same set of statements, the end of the agreeing–disagreeing spectrum that is generally considered to be more socially

<sup>10</sup>For each statement, the end of the agreeing–disagreeing spectrum that students perceived to be more socially acceptable was determined in an exploratory survey.

acceptable at UCSB and UCSD always coincides with the end of the spectrum that is closer to the average position of students in the Private treatment of the Encoding experiment who self-identify as Democrats.

*Empirical Strategy.*—The empirical analysis in this section relies almost exclusively on a simple comparison of means, which I carry out in the context of the following regression framework. Let  $Y_i$  denote an outcome variable. Let  $T_i \in \{0, 1\}$  be an indicator for being assigned to the Public treatment, and  $\varepsilon_i$  an idiosyncratic error term. I estimate the following regression equation:

$$(1) \quad Y_i = \alpha + \beta T_i + \varepsilon_i.$$

The coefficient of interest is  $\beta$ , which measures the average treatment effect of being assigned to the Public treatment compared to being assigned to the Private treatment. I use robust standard errors in all regressions.

### C. Results

In order to study whether social image concerns drive a wedge between the sensitive political attitudes that students at UCSB and UCSD report in private and in public, I estimate the average treatment effect of being assigned to the Public treatment on the attitudes reported by participants in the Encoding experiment. Let  $Y_i$  in equation (1) denote participant  $i$ 's level of agreement with one of the statements or one of the indexes. When estimating equation (1), I normalize each outcome variable  $Y_i$  so that the standard deviation of answers in the Private treatment equals one and the mean equals zero.

The results, obtained by pooling the three waves of the Encoding experiment together and shown in Figure 1, paint a consistent picture: being assigned to the Public treatment induces students to skew their answers to the sensitive questions in the direction that is generally perceived to be more socially acceptable on campus. Specifically, the point estimates of the average treatment effect on all of the sensitive statements are positive, and the one on the index of sensitive attitudes is positive (0.13 standard deviation units) and significant at the 1 percent level.<sup>11</sup> In contrast, the point estimates on the placebo statements do not exhibit a consistent pattern, and the one on the index of placebo attitudes is close to zero and statistically insignificant.

Table 2 reiterates the results obtained by pooling the three waves of the Encoding experiment and then presents the results of each individual wave. As shown in panels B, C, and D of Table 2, the results are highly heterogeneous across survey waves, arguably due to the disruptions to campus life caused by the COVID-19 pandemic.<sup>12</sup> Specifically, panels B and D of Table 2 show that the waves of data collection that occurred either before the COVID-19 pandemic (November 2019) or well after its peak (May 2023) yield results that support the experimental hypothesis, namely that,

<sup>11</sup> I provide a benchmark for the effect size after discussing the dimensions of heterogeneity that drive the treatment effects in the third wave of the Encoding experiment.

<sup>12</sup> Online Appendix Figure A1 shows the evolution of COVID-19 cases in the Santa Barbara county, where I ran the first two waves of the Encoding experiment.

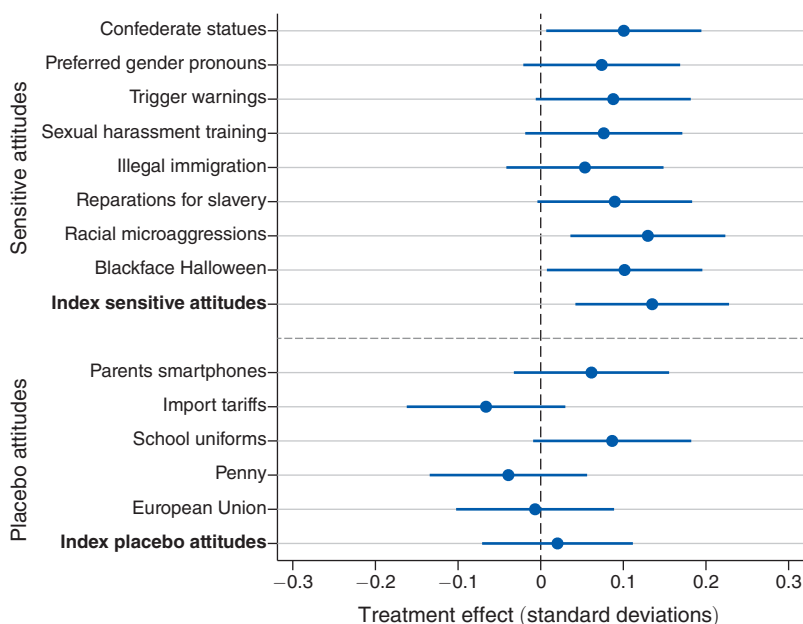


FIGURE 1. AVERAGE TREATMENT EFFECTS: SENSITIVE AND PLACEBO ATTITUDES

*Notes:* This figure presents average treatment effects of being assigned to the Public treatment using equation (1) from Section II. The results are obtained from the dataset that pools all three waves of the Encoding experiment. The figure includes the eight sensitive and five placebo statements that were common to all three waves of the Encoding experiment. The students' reported levels of agreement with the sensitive and placebo statements are oriented in such a way that larger numbers always correspond to views that are generally perceived to be more socially acceptable at UCSB and UCSD. The index of sensitive (placebo) attitudes is calculated by taking, for each participant, a simple average of the participant's answers to the sensitive (placebo) questions listed in the figure, with the answers already reoriented. All variables are normalized so that the distribution of answers of participants in the Private treatment has a mean of zero and a standard deviation of one. The normalization is achieved by first subtracting from a variable the average value of the variable among participants in the Private treatment and then dividing the result by the standard deviation of the variable among participants in the Private treatment. Error bars reflect 95 percent confidence intervals.

when assigned to the Public treatment, students skew their answers to the sensitive questions in the direction that is generally perceived to be more socially acceptable on campus. Conversely, panel C of Table 2 shows that the wave of data collection that occurred close to the peak of the COVID-19 pandemic (May 2022) yielded a null result on both the index of sensitive and the index of placebo attitudes. The pattern of heterogeneity in treatment effects across the three survey waves is consistent with the idea that the extended period of remote instruction (19 months) and the other restrictions at UCSB aimed at mitigating the COVID-19 pandemic dampened the extent to which students cared about their social image vis-à-vis their peers.

Since the Decoding experiment asked UCSD students to forecast the results of only the third wave of the Encoding experiment, the rest of this section highlights additional results from that wave. Online Appendix Figure A2 shows that the disaggregated results from wave three of the Encoding experiment are in line with the results on the pooled sample. Specifically, the sensitive statements uniformly exhibit positive treatment effects, and the coefficient on the index of sensitive attitudes is

TABLE 2—AVERAGE TREATMENT EFFECTS ON THE INDEXES OF SENSITIVE AND PLACEBO ATTITUDES

|                                                   | Index sensitive attitudes<br>(1) | Index placebo attitudes<br>(2) |
|---------------------------------------------------|----------------------------------|--------------------------------|
| <i>Panel A. All waves together</i>                |                                  |                                |
| Average treatment effect                          | 0.13<br>(0.05)                   | 0.02<br>(0.05)                 |
| Observations                                      | 1,699                            | 1,699                          |
| <i>Panel B. Wave 1 (pre-COVID-19 pandemic)</i>    |                                  |                                |
| Average treatment effect                          | 0.20<br>(0.11)                   | −0.02<br>(0.10)                |
| Observations                                      | 320                              | 320                            |
| <i>Panel C. Wave 2 (during COVID-19 pandemic)</i> |                                  |                                |
| Average treatment effect                          | 0.07<br>(0.09)                   | 0.01<br>(0.08)                 |
| Observations                                      | 551                              | 551                            |
| <i>Panel D. Wave 3 (post-COVID-19 pandemic)</i>   |                                  |                                |
| Average treatment effect                          | 0.15<br>(0.07)                   | 0.04<br>(0.07)                 |
| Observations                                      | 828                              | 828                            |

*Notes:* This table presents average treatment effects of being assigned to the Public treatment using equation (1) from Section II. Panel A presents the results obtained from the dataset that pools all three waves of the Encoding experiment; panels B, C, and D present the results of the individual waves of the experiment. The dependent variables are the normalized indexes of the level of agreement with the sensitive and placebo statements that are common across the three survey waves. The index of sensitive (placebo) attitudes is calculated by taking, for each participant, a simple average of the participant's answers to the sensitive (placebo) questions that are common across the three survey waves, with the answers oriented in such a way that larger numbers always correspond to views that are generally perceived to be more socially acceptable at UCSB and UCSD. The normalization is achieved by first subtracting from an index the average value of the index among participants in the Private treatment and then dividing the result by the standard deviation of the index among participants in the Private treatment. Standard errors are in parentheses.

positive and significant. Conversely, the placebo statements exhibit less clear a pattern, and the index of placebo attitudes is not significantly different from zero.

Figure 2 describes treatment effect heterogeneity in the third wave of the Encoding experiment across a host of student characteristics. Specifically, in the pre-analysis plan, I specified the following moderators for the heterogeneous treatment effect analysis: gender, race/ethnicity, political affiliation, year in school, and whether the subjects' major is in the humanities or in the sciences. Figure 2 presents estimates of the interaction term between being assigned to the Public treatment and each moderator. As shown in Figure 2, the treatment effects differ widely depending on the participants' self-reported political affiliation. Specifically, students who self-identify as Independents or Republicans exhibit significantly larger treatment effects than students who self-identify as Democrats.<sup>13</sup> No other significant dimension of heterogeneity emerges from the analysis.

<sup>13</sup>The heterogeneity in treatment effect by political ideology is in line with the model of tolerance and compromises on social networks by Genicot (2022). Furthermore, it is in line with the empirical findings in Greenberg and Stone (2024).

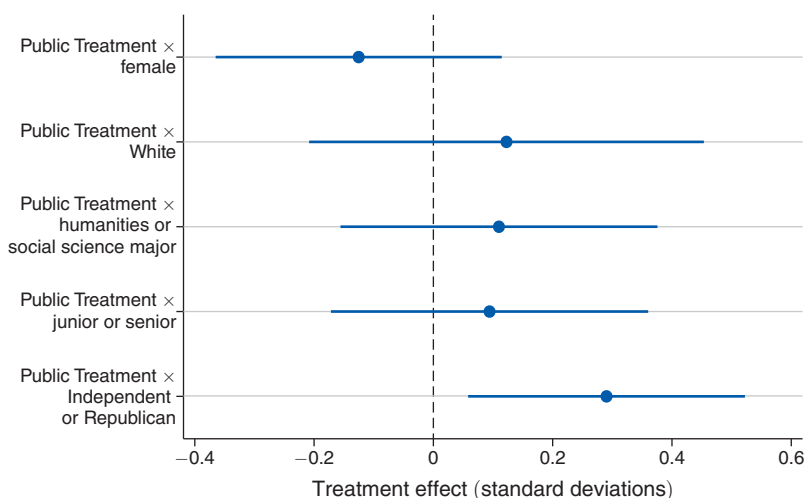


FIGURE 2. HETEROGENEOUS TREATMENT EFFECTS ON THE INDEX OF SENSITIVE ATTITUDES

*Notes:* This figure presents, based on data from wave three of the Encoding experiment, the coefficients on the interaction terms in the following regression equation  $Y_i = \alpha + \beta T_i + \delta M_i + \gamma T_i \times M_i + \varepsilon_i$ , where  $Y_i$  denotes the standardized index of sensitive attitudes,  $T_i$  is an indicator that takes value 1 if individual  $i$  is assigned to the Public treatment,  $M_i$  is an indicator for the category of the binary moderator participant  $i$  belongs to, and  $\varepsilon_i$  is an idiosyncratic error term. The index of sensitive attitudes is calculated by taking, for each participant, a simple average of the participant's answers to the ten sensitive questions in wave three of the Encoding experiment, with the answers oriented in such a way that larger numbers always correspond to views that are generally perceived to be more socially acceptable at UCSB. The normalization is achieved by first subtracting from an index the average value of the index among participants in the Private treatment and then dividing the result by the standard deviation of the index among participants in the Private treatment. Error bars reflect 95 percent confidence intervals.

The heterogeneous treatment effects on political affiliation, beside being important for the study of information loss as discussed in the next two sections, can help build intuition about the overall magnitude of the treatment effects in the Encoding experiment and may inform the debate about the perceived lack of ideological diversity on college campuses (Inbar and Lammers 2012). I already discussed how, for each of the sensitive statements, the end of the agreeing–disagreeing spectrum that is generally considered to be more socially acceptable at UCSB and UCSD always corresponds to the end of the spectrum that is closer to the average position of self-identified Democrats in the Private treatment of the Encoding experiment. Therefore, assuming the Likert scale can be interpreted as linear, we can measure the ideological gap between self-reported Democrats and self-reported Independents/Republicans by calculating the absolute difference between the average value of the index of sensitive attitudes among students who self-identify as Democrats and among students who self-identify as Independents or Republicans. Doing so separately for the Private and the Public treatment shows that the ideological gap in the Private treatment is 1.4 times larger than the ideological gap in the Public treatment. Therefore, if one were to naïvely assess the degree of ideological diversity on college campus by taking public statements at face value, one would underestimate the degree of ideological diversity on campus by around 40 percent.



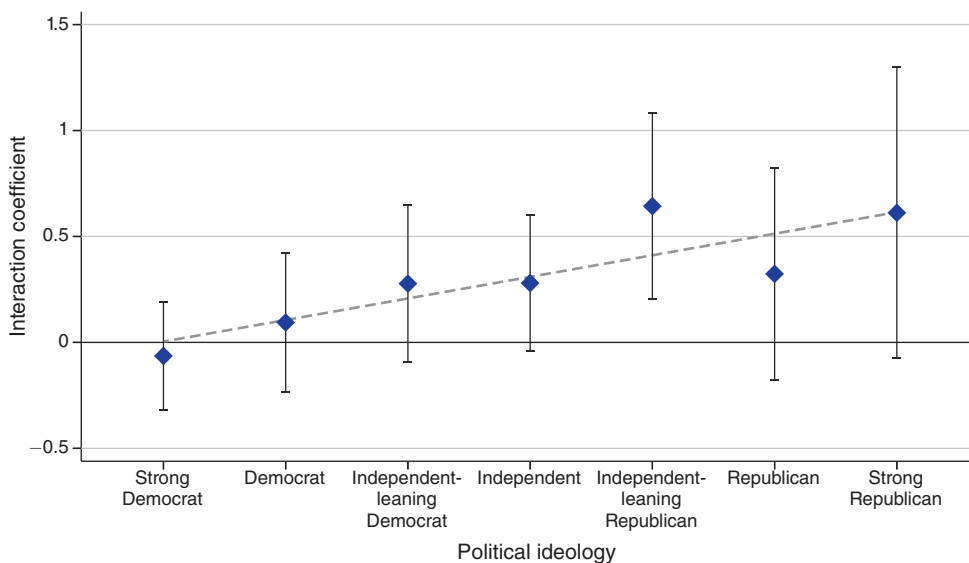


FIGURE 3. HETEROGENEOUS TREATMENT EFFECTS BY POLITICAL IDEOLOGY

*Notes:* This figure presents, based on data from wave three of the Encoding experiment, the coefficient on the interaction term in the following regression equation  $Y_i = \sum_{j=1}^7 \delta_j P_i^j + \sum_{j=1}^7 \gamma_j T_i \times P_i^j + \varepsilon_i$ , where  $Y_i$  denotes the standardized index of sensitive attitudes,  $T_i$  is an indicator that takes value 1 if individual  $i$  is assigned to the Public treatment,  $\{P_i^j\}_{j=1}^7$  is a set of indicator variables for the political ideology category  $j \in \{1, \dots, 7\}$  that individual  $i$  reported belonging to, and  $\varepsilon_i$  is an idiosyncratic error term. The index of sensitive attitudes is calculated by taking, for each participant, a simple average of the participant's answers to the ten sensitive questions in wave three of the Encoding experiment, with the answers oriented in such a way that larger numbers always correspond to views that are generally perceived to be more socially acceptable at UCSD. The normalization is achieved by first subtracting from the index the average value of the index among participants in the Private treatment and then dividing the result by the standard deviation of the index among participants in the Private treatment. Error bars reflect 95 percent confidence intervals. The dashed line in the figure is obtained from a regression similar to the one above but in which political affiliation enters linearly rather than as a set of indicators. The coefficient on the interaction between the linear political affiliation variable and being assigned to the Public treatment is 0.1, and it is significant at the 1 percent level ( $p$ -value: 0.004).

In order to dig deeper into the heterogeneous treatment effects on self-reported political affiliation, Figure 3 presents heterogeneous treatment effects coefficients for each of the seven political categories that participants in the third wave of the Encoding experiment could identify with. The figure shows that the treatment effects of being assigned to the Public treatment become steadily larger the more closely one identifies with the Republican end of the political spectrum.

*Robustness.*—As robustness checks, online Appendix Table A6 shows that the results on the index of sensitive attitudes are not driven by any single statement, and online Appendix Table A7 shows that they are robust to relaxing the assumption that the Likert scale can be interpreted as linear. Online Appendix Table A8 provides regression results for all the sensitive statements in both normalized (standard deviation) units and unnormalized (original) units. The table also provides unadjusted  $p$ -values and “sharpened” false discovery rate (FDR)-adjusted  $q$ -values following the procedure of Benjamini, Krieger, and Yekutieli (2006), as outlined by Anderson

(2008).<sup>14</sup> Lastly, online Appendix D.2.1 presents a robustness check suggesting that the heterogeneous treatment effects from the previous section are not driven by ceiling effects.<sup>15</sup>

### III. Information Loss: Encoding Process

Do the equilibrium distortions caused by social image lead to information loss in the encoding process? In other words, do they lead to information loss even from the perspective of a sophisticated audience that perfectly understands the ways in which social image distorts the students' publicly stated attitudes? In this section, I rely on data from the third wave of the Encoding experiment to show that the answer is affirmative and that the magnitude of information loss can be substantial even for a sophisticated audience.

In order to study the degree of informativeness of the answers given by participants in the Encoding experiment, it is useful to adopt a canonical measure of informativeness that satisfies at least two desiderata. First, the measure has to be theoretically grounded in the sense of being related to the signaling model developed in online Appendix A. Second, the measure has to be cardinal so as to be able to capture the magnitude of information loss.

The canonical information-theoretic notion of *mutual information* satisfies both desiderata.<sup>16</sup> First, it follows naturally from the model in online Appendix A because it is implied by Blackwell informativeness via the data-processing inequality (Cover and Thomas 2006). Therefore, if social image concerns induce distortions that diminish the informativeness of equilibrium statements according to the Blackwell partial order as I show in online Appendix A, they also diminish the equilibrium statements according to the mutual information order. Second, mutual information has cardinal value.

Since the students' private opinions on the sensitive issues covered in the Encoding experiment are not directly observable, I study the mutual information between the students' public statements and a host of random variables that I expect to be correlated with the students' private opinions.<sup>17</sup> Specifically, I consider the relationship between a participant's index of sensitive attitudes and whether the participant

<sup>14</sup>The unadjusted  $p$ -values are appropriate for readers with a priori interest in one specific outcome. The FDR-adjusted  $q$ -values for the individual outcomes limit the expected proportion of false rejections of null hypotheses across all sensitive statements. The sharpened FDR-adjusted  $q$ -values are less conservative than the unadjusted  $p$ -values for  $p$ -values greater than about 0.15, and more conservative for unadjusted  $p$ -values less than that. To alleviate concerns with multiple hypothesis testing, the pre-analysis plan specified the indexes as the main outcome variables, rather than the individual statements (Allcott et al 2020).

<sup>15</sup>One can imagine a world in which the treatment effects on Democrats and Republicans are in principle identical but in which ceiling effects mechanically constrain the treatment effects on Democrats and not on Republicans.

<sup>16</sup>Mutual information is defined as the Kullback-Leibler divergence between the joint distribution of two random variables and the product of their marginal distributions. It measures the degree of dependency of two random variables; specifically, it measures the reduction in entropy about a certain random variable that comes from observing the realization of another (Cover and Thomas 2006).

<sup>17</sup>The students' private opinions are unobservable by definition: in both the Private and the Public treatments, I can only observe the students' reports about their private opinions. One could in principle impose the assumption that the answers of students assigned to the Private treatment reflect the students' private opinions. The analysis in this section is designed to be robust to violations of that assumption. In other words, the analysis in this section is designed to accommodate the possibility that students assigned to the Private treatment may not truthfully report their private opinions, which would happen, for instance, if they worried their answers would not be kept fully confidential.

self-identified as a Democrat or as an Independent/Republican. Furthermore, I consider the relationship between a participant's level of agreement with the statement about transgender athletes (statement 4 in Table 1) and the amount the participant chose to donate to Athlete Ally.

As a preliminary step, online Appendix Table A9 shows that the target random variables—self-identifying as a Democrat or as an Independent/Republican and the amount donated to Athlete Ally—are not affected by treatment status. This is consistent with the fact that such variables were elicited before treatment assignment. Therefore, information loss about the target random variables is unlikely to be due to those variables varying with treatment, but rather to the levels of agreement with the sensitive statements varying with treatment.

Figure 4 shows the mutual information between (i) a participant's index of sensitive attitudes and whether the participant self-identifies as a Democrat or as an Independent/Republican and (ii) a participant's reported level of agreement with the statement about transgender athletes and the participant's donation to Athlete Ally. The results show that, for both sets of variables, mutual information is larger in the Private treatment than in the Public treatment. According to the mutual information measure, the statements of participants in the Public treatment are 30 to 50 percent less informative about the target random variables than the statements of participants in the Private treatment even from the perspective of a sophisticated audience. Another way of phrasing the statement above is that, according to the mutual information measure, the amount of information about the target random variables that is lost in the encoding process is between 30 percent and 50 percent.

As a robustness check, online Appendix E.1 shows that the results are unaltered if, rather than mutual information, one employs other canonical notions of informativeness, such as measures of classifier performance often used in the machine learning literature.

#### IV. Information Loss: Decoding Process

The previous section studied information loss in the process of encoding. In other words, it studied whether the signal structures generated by students assigned to the Private treatment of wave three of the Encoding experiment are more informative than the signal structures generated by students assigned to the Public treatment. As discussed, one can think of information loss in the encoding process as the amount of information that would be lost to a sophisticated audience who recognize the ways in which social image concerns distort the attitudes that students display in public and who adjust their inferences accordingly. If the natural audience in the environment, namely, other students, are less sophisticated in understanding the ways in which social image distorts the attitudes portrayed in public by their peers, they might extract even less information from public statements. In order to study whether audience naïveté leads to additional information loss, I employed a second experiment: the Decoding experiment.

The design of the Decoding experiment had two core objectives. First, to test whether students neglect the effects of social image when it is not salient on their minds. Second, to study how students interpret their peers' public statements when social image is salient on their minds. The two objectives are motivated by a model

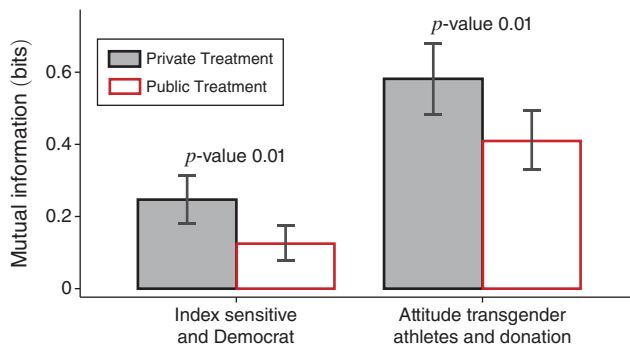


FIGURE 4. MUTUAL INFORMATION

*Notes:* This figure presents estimates of mutual information (in bits) for different duples of variables and separately for participants in the Private and the Public treatments of wave three of the Encoding experiment. The first set of columns shows estimates of the mutual information between a participant's index of sensitive attitudes, rounded to the nearest digit, and whether the participant self-identified as a Democrat. The index of sensitive attitudes is calculated by taking, for each participant, a simple average of the participant's answers to the ten sensitive questions in wave three of the Encoding experiment, with the answers oriented in such a way that larger numbers always correspond to views that are generally perceived to be more socially acceptable at UCSD. Self-identifying as a Democrat is defined as selecting the answer options of "Strong Democrat" or "Democrat" to the political affiliation questions, whose other answer options were "Independent-Leaning Democrat," "Independent," "Independent-Leaning Republican," "Republican," and "Strong Republican." The second set of columns shows estimates of the mutual information between a participant's reported level of agreement with the statement about transgender athletes (statement 4 in Table 1) and the participant's donation to Athlete Ally. Error bars reflect 95 percent confidence intervals.

of cognition where, as described in Daniel Kahneman's *Thinking, Fast and Slow* (Kahneman 2012), individuals rely on two systems of thinking: System 1, which is fast, automatic, and often influenced by heuristics, and System 2, which is slower, more deliberate, and analytical. This model suggests that when social image is not salient, students may rely on System 1 and overlook social image effects. However, when social image is made salient, System 2 may be more engaged, leading to a deeper consideration of the motivations behind public statements. An overview of the design of the Decoding experiment is given in Figure 5.

#### A. Decoding Experiment: Design

In November and December 2023, I recruited a new set of students from the University of California San Diego to complete a brief online survey. Before being directed to the consent form, participants were asked to complete a brief prescreen questionnaire similar to the one administered to subjects in the Encoding experiment. A student was considered eligible to participate in the experiment if she reported being an undergraduate at UCSD.<sup>18</sup>

After completing the prescreen survey and consenting to take part in the experiment, subjects were informed that their task would be to forecast the answers of a group of UCSD undergraduate students who participated in an experiment (wave three of the

<sup>18</sup>The full instructions for the experiment are available in online Appendix I.

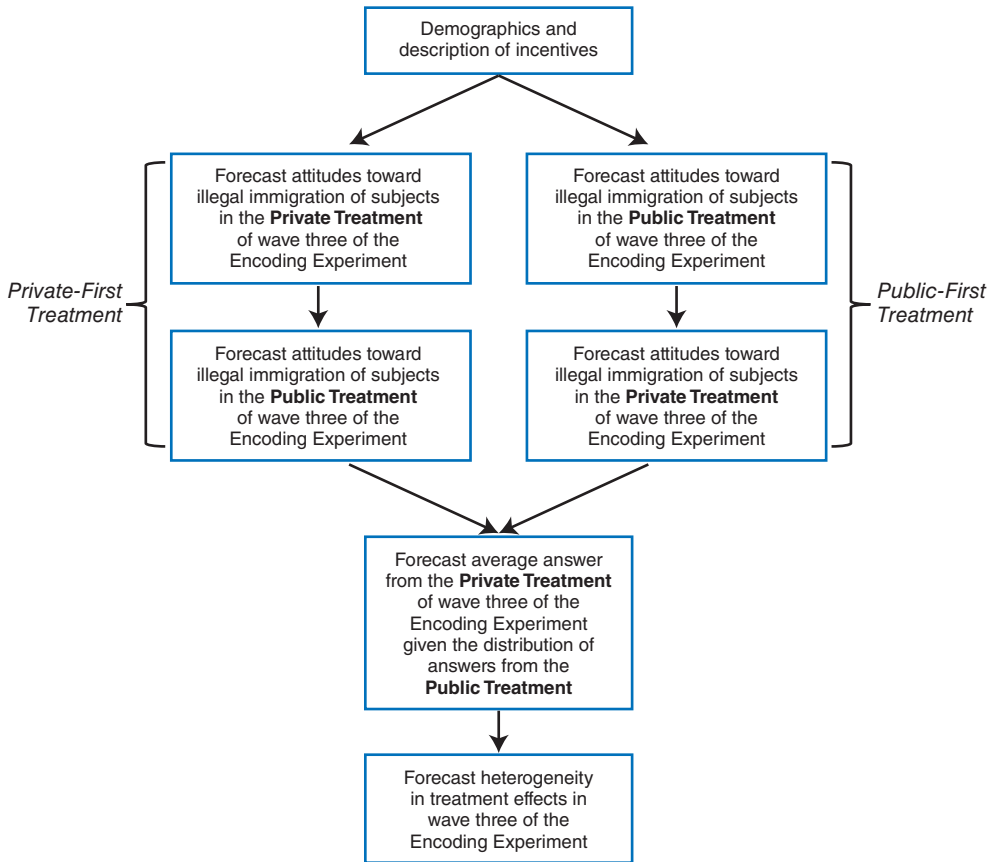


FIGURE 5. OVERVIEW OF THE DESIGN OF THE DECODING EXPERIMENT

*Notes:* This figure provides an overview of the main design features of the Decoding experiment. The full instructions for the experiment are available in online Appendix I.

Encoding experiment) run a few months earlier. Subjects were instructed to predict the responses of participants exclusively from wave three of the Encoding experiment, as equilibrium reasoning is more likely when speakers and their audience share the same environment.<sup>19</sup> Subjects were informed that each of them would have a chance to earn a \$10 bonus payment with a probability increasing in the accuracy of their forecasts.

The details of the lottery system used to allocate the bonus payments to participants are as follows. At the end of the survey, each participant was entered into a lottery that paid either \$10 or \$0. The lotteries were independent across participants. For each participant, the probability with which the lottery paid \$10 was a function of the average accuracy of the participant's forecasts across all the questions in the experiment. Specifically, the more accurate the participant's forecasts, the higher

<sup>19</sup>I also ran a version of the Decoding experiment at UCSB in 2020 with a slightly less rich design; the results are presented in online Appendix G.

the probability of the \$10 outcome. Such payment system has the property of being incentive compatible irrespective of the participant's degree of risk aversion.<sup>20</sup>

Participants were informed of the details of the payment system, including the mathematical formulas used to generate the probability of being awarded the \$10 bonus payments, and were told that it was in their best interest to give their most accurate guess to each question. Furthermore, in order to familiarize participants in the Decoding experiment with the characteristics of the sample of students whose behavior they had to forecast, they were given information about the demographic characteristics of the sample of subjects who took part in wave three of the Encoding experiment.

Participants were then randomized into one of two treatments: a Private-First or a Public-First treatment. The purpose of such randomization was to vary the salience of social image across two sets of questions and to study the effects of salience on the participants' forecasts. Specifically, when answering the first set of questions, participants were made aware of the existence of *only one* of the treatments of the Encoding experiment; conversely, when answering the second set of questions, participants were made aware of the existence of *both* treatments of the Encoding experiment. Learning that the Encoding experiment had two treatments and being shown the instructions for both was supposed to draw participants' attention to social image, thus making social image more salient in the second set of questions than in the first. For this reason, I conjectured that participants in the Decoding experiment would be more likely to take into account the effects of social image in their second set of forecasts than in their first.

The implementation of the treatment variation described above was as follows. Participants assigned to the Private-First treatment were shown the instructions of subjects assigned to the Private treatment in wave three of the Encoding experiment and were asked to forecast the fraction of Democrats and of Independents/Republicans in that treatment of the Encoding experiment who agreed with the statement about illegal immigration (statement 7 in Table 1).<sup>21</sup> Participants assigned to the Public-First treatment were asked to make the same forecasts but for subjects in the Public treatment of wave three of the Encoding experiment. This way, when making their first set of forecasts, participants in the Decoding experiment were aware of the existence of only one of the treatments of the Encoding experiment. After making their first set of forecasts, participants in the Decoding experiment were informed of the existence of the other treatment of the Encoding experiment and were asked to make an analogous set of forecasts, but for that treatment. Therefore, by their second set of forecasts, participants in the Decoding experiment were aware of both treatments of the Encoding experiment, which was supposed to increase the salience of social image in the participants' minds.

In the next part of the Decoding experiment, participants were shown, for each of the 10 sensitive and 5 placebo statements, the distribution of answers of subjects

<sup>20</sup>The probability distribution over payments generated by the procedure above is equivalent to the one generated by a two-step procedure in which, first, a question is randomly selected for each participant, and, second, each participant is awarded a \$10 bonus with a probability that depends on the accuracy of her answer to the selected question.

<sup>21</sup>Agreeing with the statement about illegal immigration was defined as answering a number strictly larger than 5 on the 0–10 scale used in the Encoding experiment.



in the Public treatment of wave three of the Encoding experiment and were asked to forecast the average answer of subjects in the Private treatment. The aim of this exercise was to mimic the key inference problem that a student faces upon hearing a peer make a “politically correct” statement in public, namely, to try to recover the peer’s private opinion from her public statement. Participants in the Decoding experiment reported their forecasts on a 0–10 slider that allowed them to select numbers with up to one decimal place. The order of the statements was randomized at the participant level.

In the last part of the Decoding experiment, participants were asked to forecast the dimensions of heterogeneity that drive the average treatment effects in wave three of the Encoding experiment. Specifically, participants were informed that, for 10 of the statements, the experimenter had hypothesized that the average answers of subjects in the Private and the Public treatment of the Encoding experiment would differ systematically. After being shown those 10 statements (the sensitive statements) participants were asked to forecast whether the effects of being assigned to the Public treatment compared to the Private treatment were, on average across the 10 statements, larger for subjects belonging to a certain demographic group than for subjects belonging to the complementary demographic group. The questions were multiple-choice: subjects could answer that the treatment effects were significantly larger for the first demographic group, significantly larger for the second demographic group, or that the treatment effects were not significantly different from one another.<sup>22</sup> The demographic groups were the same as the ones prespecified for the heterogeneity analysis in the Encoding experiment, namely gender, race/ethnicity, political affiliation, year in school, and whether the student’s major was in the humanities or in the sciences. The order of the questions was randomized at the subject level, and participants were also asked to report their degree of confidence in their forecasts.

Upon completing the survey, a subset of participants was shown the aggregate-level answers of subjects in the Private treatment of wave three of the Encoding experiment and was redirected to a spreadsheet that contained the anonymized individual-level answers of subjects in the Public treatment of wave three of the Encoding experiment. Bonus payments were calculated and disbursed a few days after the end of the survey.

*Outcome Variables.*—The first set of outcome variables consists of the subjects’ forecasts of the answers of self-identified Democrats and self-identified Independents/Republicans in wave three of the Encoding experiment to the question about illegal immigration. As described above, each participant in the Decoding experiment made forecasts about both the Private and the Public treatment of the Encoding experiment, but the order of those forecasts was randomized. This way, comparing the first set of forecasts of participants in the Private-First treatment to those of participants in the Public-First treatment of the Decoding experiment allows me to study whether students take the effects of social image into account when social image is arguably not very salient. Conversely, comparing the second set of

<sup>22</sup> Subjects were informed of what constituted a statistically significant difference in the context of the Encoding experiment.

forecasts of participants in the Private-First treatment to those of participants in the Public-First treatment of the Decoding experiment allows me to study whether participants take the effects of social image into account when social image is arguably more salient.

The second set of outcome variables considers the forecasts by participants in the Decoding experiment of the average private levels of agreement of participants in wave three of the Encoding experiment. The third outcome variable captures the subjects' forecasts of the heterogeneous treatment effects in wave three of the Encoding experiment.

### B. Results

Figure 6 shows that subjects neglect the effects of social image when social image is not made salient. Specifically, Figure 6 contains three panels. The first panel shows, as a benchmark, results from the Encoding experiment. In particular, it shows that (i) self-identified Democrats are statistically as likely to agree with the statement about illegal immigration in the Private treatment as in the Public treatment, and (ii) Independents and Republicans are significantly more likely to agree with the statement about illegal immigration in the Private treatment than in the Public treatment. The pattern of results is consistent with the previous finding that Independents and Republicans in the Encoding experiment tend to skew their public answers in the liberal direction.

The second panel of Figure 6 shows the first set of forecasts of participants in the Decoding experiment, namely, the ones made when social image is arguably not salient. Consistent with social image not being salient in the subjects' minds, the forecasts of subjects who were asked to predict the answers of participants in the Private treatment of the Encoding experiment are statistically indistinguishable from the forecasts of subjects who were asked to predict the answers of participants in the Public treatment of the Encoding experiment.

In contrast, the third panel of Figure 6 shows the forecasts of participants in the Decoding experiment when social image was made more salient. Specifically, upon learning that the Encoding experiment had both a Private and a Public treatment and being shown the instructions of both, participants in the Decoding experiment predicted students assigned to the Public treatment of the Encoding experiment to be less likely to agree with the statement about illegal immigration than students assigned to the Private treatment. Therefore, the first headline result of the Decoding experiment is that, when making their forecasts, students appear to take the effects of social image into account only when it is made salient.

The next set of results show that, conditional on social image being salient, students are quite proficient at interpreting their peers' public statements, though they tend to overestimate the effects of social image. Figure 7 displays, for each sensitive statement in wave three of the Encoding experiment, the average level of agreement of participants in the Public treatment, the average level of agreement of participants in the Private treatment, and the mean forecast, by participants in the Decoding experiment, of the average private level of agreement from wave three of the Encoding experiment. The figure shows that, for all the sensitive statements, both the average level of agreement of participants in the Private treatment of the

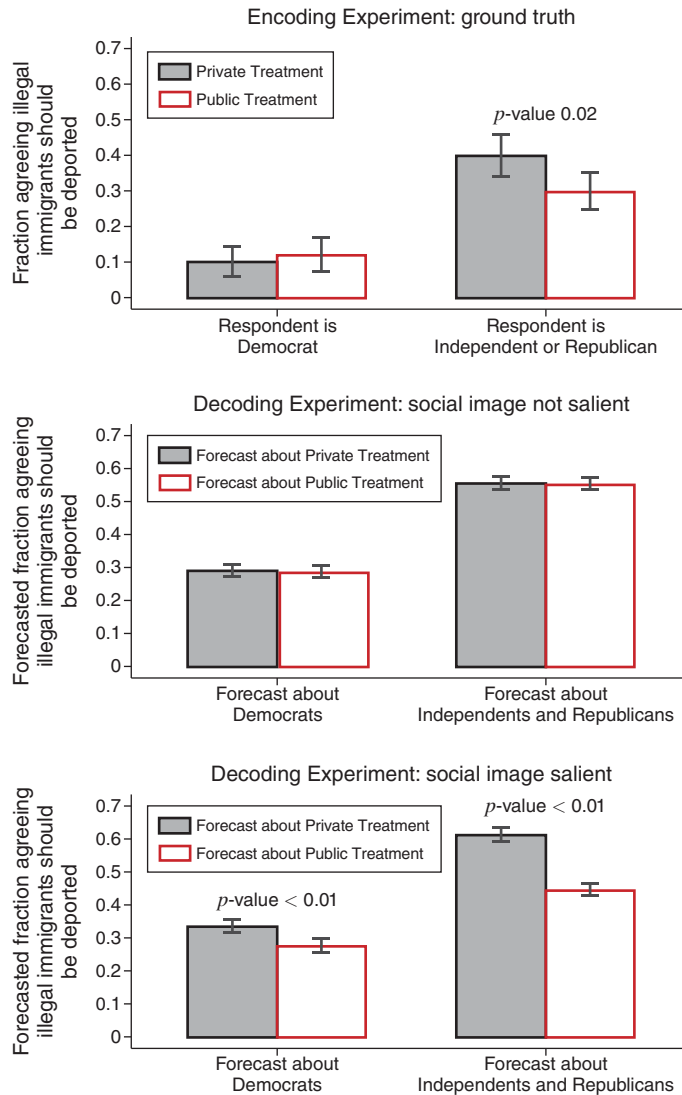


FIGURE 6. EFFECTS OF SOCIAL IMAGE SALIENCE ON THE STUDENTS' FORECASTS

*Notes:* The first panel of the figure shows the fraction of individuals in wave three of the Encoding experiment who reported agreeing with the statement about illegal immigration (statement 7 in Table 1) by treatment status and self-reported political affiliation. Agreeing with the statement was defined as selecting a number strictly larger than 5 on the 0–10 scale on which students in the Encoding experiment were asked to indicate their level of agreement with the statement. The second panel shows the average forecasts of participants in the Decoding experiment before being told that the Encoding experiment had two treatments and after being shown the instructions for only one of the treatments. The third panel shows the average forecasts of participants in the Decoding experiment after being told that the Encoding experiment had two treatments and after being shown the instructions for both treatments. Error bars reflect 95 percent confidence intervals.

Encoding experiment and the mean forecast by participants in the Decoding experiment are always below the average level of agreement of participants in the Public treatment of the Encoding experiment. Therefore, when social image is salient, UCSD students have an accurate perception of the direction in which their peers skew their public answers.

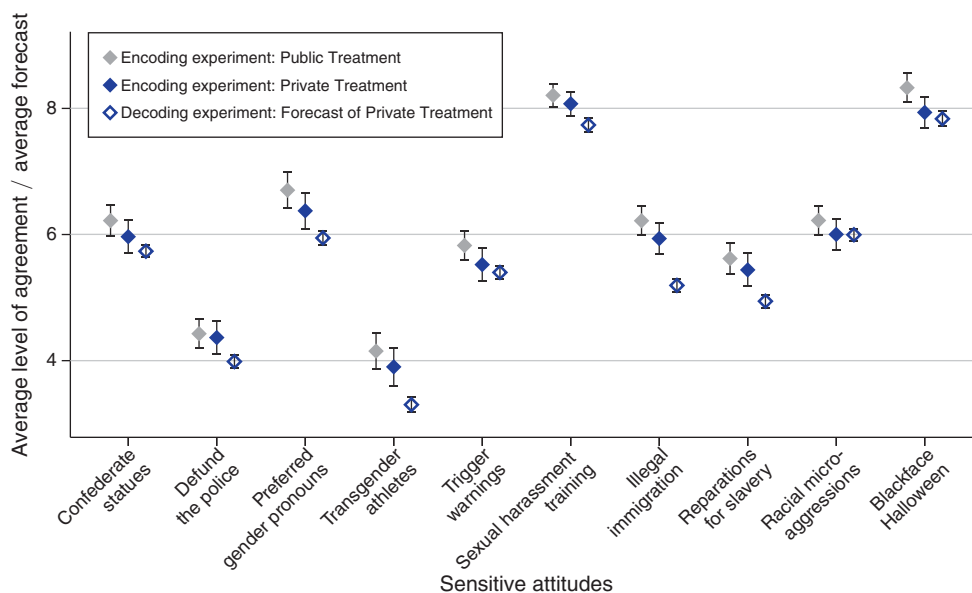


FIGURE 7. ACTUAL AVERAGE LEVELS OF AGREEMENT VERSUS FORECASTS: SENSITIVE STATEMENTS

*Notes:* This figure shows, for each sensitive statement, the average level of agreement of participants in the Public treatment of wave three of the Encoding experiment, the average level of agreement of participants in the Private treatment of wave three of the Encoding experiment, and the mean forecast, by participants in the Decoding experiment, of the average private level of agreement from wave three of the Encoding experiment. All answers are oriented in such a way that larger numbers always correspond to views that are generally perceived to be more socially acceptable at UCSB. Error bars reflect 95 percent confidence intervals.

Figure 7 also shows that participants in the Decoding experiment slightly overestimate the degree to which social image concerns induce their peers to distort their answers to the sensitive statements. Specifically, the point estimate of the mean forecast by participants in the Decoding experiment is virtually always below the average private level of agreement from wave three of the Encoding experiment. Since in Figure 7 all answers are oriented in such a way that larger numbers always correspond to views that are perceived to be more socially acceptable at UCSB, finding that the point estimate of the mean forecast is virtually always below the point estimate for the actual average level of agreement with a statement is an indication that participants in the Decoding experiment believe that the effects of social image on the public attitudes portrayed by their peers are stronger than they actually are. In fact, when pooling results across the ten sensitive statements, the difference between the average level of agreement of participants in the Private treatment of wave three of the Encoding experiment and the mean forecast of participants in the Decoding experiment is significant at the 5 percent level. Online Appendix Figure A6 presents an analogous graph, but for the placebo statements. The figure does not exhibit a consistent pattern of results. Furthermore, when pooling results across the five placebo statements, there is no significant difference between the average level of agreement of participants in the Private treatment of wave three of the Encoding experiment and the mean

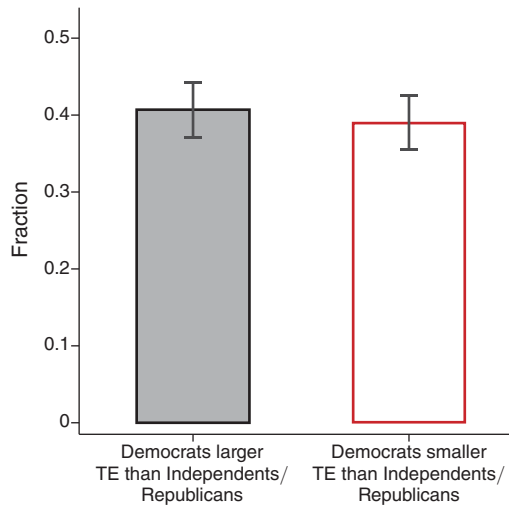


FIGURE 8. FORECASTED HETEROGENEITY IN TREATMENT EFFECTS BY POLITICAL IDEOLOGY

*Notes:* The figure above shows the distribution of answers of participants in the Decoding experiment to the multiple-choice questions about heterogeneity by political ideology. The first bar represents the fraction of participants in the Decoding experiment who answered that self-identified Democrats exhibited significantly larger treatment effects than self-identified Independents or Republicans. The second bar represents the fraction of participants in the Decoding experiment who answered that self-identified Democrats exhibited significantly smaller treatment effects than self-identified Independents or Republicans. The fraction of participants in the Decoding experiment who answered that self-identified Democrats and self-identified Independents/Republicans exhibited treatment effects that are not significantly different from one another is omitted and can be recovered from the two bars in the figure. Error bars reflect 95 percent confidence intervals.

forecast of participants in the Decoding experiment ( $p$ -value 0.62). Therefore, for the sensitive but not for the placebo statements, UCSB students overestimate the degree to which social image affects the attitudes that their peers portray in public.

The last figure, Figure 8, shows that students do not understand the main dimension of heterogeneity that drives the treatment effects in wave three of the Encoding experiment. Specifically, Figure 8 presents a histogram of the subjects' answers to a question asking whether the effects of being assigned to the Public treatment compared to the Private treatment are, on average across the ten sensitive statements, significantly larger for self-identified Democrats, significantly larger for self-identified Independents/Republicans, or not significantly different across the two groups. As shown in Figure 8, the fraction of participants who report believing that self-identified Democrats exhibit larger treatment effects than self-identified Independents/Republicans (41 percent) is statistically indistinguishable from the fraction of participants who report the opposite (39 percent).

Overall, the results of the Decoding experiment can be summarized as follows. First, students neglect the effects of social image on their peers' public statements unless social image is made salient. Second, when social image is made salient, students exhibit a qualitatively accurate understanding of the ways in which social image distorts their peers' public answers to the sensitive questions, but they overestimate the magnitude of such distortions. Third, students fail to recognize that the effects of social image are highly heterogeneous by political ideology, thus

misunderstanding a crucial dimension of how social image concerns operate on their campus.

## V. Conclusion

An important argument in the cultural debate about political correctness is that people may feel pressure to publicly espouse views on a set of sensitive political topics that they may not privately hold, and that such misrepresentations may impoverish the quality of public discourse. This paper provides a formalization of the argument and tests it in an environment that is often at the center of the debate, namely, that of college campuses. Broadly speaking, the results lend empirical support to the argument: students indeed skew their public utterances in the direction that is perceived to be more socially acceptable on campus, and such distortions cause public statements to be less informative than private statements according to a host of canonical measures of informativeness. Information transmission is further impaired because students tend to neglect the role of social image on their peers' public statements unless it is made salient. When the role of social image is made salient, students exhibit a nuanced, although imperfect, understanding of how social image concerns influence the attitudes portrayed in public by their peers.

This paper can be seen as a first step in a broader welfare analysis of the phenomenon of political correctness. Specifically, in environments where information transmission is desirable, the paper suggests that social image concerns around topics related to political correctness may exact a toll in terms of information transmission. Clearly, a full welfare analysis requires a proper understanding of the benefit side of political correctness as well. Developing an empirically grounded understanding of the benefits of political correctness is an interesting and important avenue for future research.

## REFERENCES

- Ackerman, Elliot, Saladin Ambar, Martin Amis, Anne Applebaum, Marie Arana, Margaret Atwood, John Banville, et al. 2020. "A Letter on Justice and Open Debate." *Harper's Magazine*, July 7. <https://harpers.org/a-letter-on-justice-and-open-debate/>.
- Allcott, Hunt, Luca Braghieri, Sarah Eichmeyer, and Matthew Gentzkow. 2020. "The Welfare Effects of Social Media." *American Economic Review*, 110 (3): 629–76.
- Alpizar, Francisco, Fredrik Carlsson, and Olof Johansson-Stenman. 2008. "Anonymity, Reciprocity, and Conformity: Evidence from Voluntary Contributions to a National Park in Costa Rica." *Journal of Public Economics* 92 (5-6): 1047–60.
- Anderson, Michael L. 2008. "Multiple Inference and Gender Differences in the Effects of Early Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects." *Journal of the American Statistical Association* 103 (484): 1481–95.
- Banerjee, Abhijit, Emily Breza, Arun G. Chandrasekhar, and Benjamin Golub. 2024. "When Less is More: Experimental Evidence on Information Delivery During India's Demonetization." *Review of Economic Studies* 91 (4): 1884–1922.
- Benjamini, Yoav, Abba M. Krieger, and Daniel Yekutieli. 2006. "Adaptive Linear Step-Up Procedures That Control the False Discovery Rate." *Biometrika* 93 (3): 491–507.
- Braghieri, Luca. 2023. "Biased Decoding and the Foundations of Communication." Unpublished.
- Braghieri, Luca. 2023a. "Political Correctness, Social Image, and Information Transmission. Encoding Experiment." AEA RCT Registry. <https://doi.org/10.1257/rct.11440-1.1>.
- Braghieri, Luca. 2023b. "Political Correctness, Social Image, and Information Transmission. Decoding Experiment." AEA RCT Registry. <https://doi.org/10.1257/rct.11981-1.0>.



- Braghieri, Luca.** 2024. *Data and Code for: "Political Correctness, Social Image, and Information Transmission."* Nashville, TN: American Economic Association; distributed by Inter-university Consortium for Political and Social Research, Ann Arbor, MI. <https://doi.org/10.3886/E208243V1>.
- Bursztyn, Leonardo, Thomas Fujiwara, and Amanda Pallais.** 2017. "'Acting Wife': Marriage Market Incentives and Labor Market Investments." *American Economic Review* 107 (11): 3288–3319.
- Bursztyn, Leonardo, Bruno Ferman, Stefano Fiorin, Martin Kanz, and Gautam Rao.** 2018. "Status Goods: Experimental Evidence from Platinum Credit Cards." *Quarterly Journal of Economics* 133 (3): 1561–95.
- Bursztyn, Leonardo, Georgy Egorov, and Stefano Fiorin.** 2020. "From Extreme to Mainstream: The Erosion of Social Norms." *American Economic Review* 110 (11): 3522–48.
- Bursztyn, Leonardo, Alessandra González, and David Yanagizawa-Drott.** 2020. "Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia." *American Economic Review* 110 (10): 2997–3029.
- Butera, Luigi, Robert Metcalfe, William Morrison, and Dmitry Taubinsky.** 2022. "Measuring the Welfare Effects of Shame and Pride." *American Economic Review* 112 (1): 122–68.
- Chandrasekhar, Arun G., Benjamin Golub, and He Yang.** 2019. "Signaling, Shame, and Silence in Social Learning." Unpublished.
- Cover, Thomas M., and Joy A. Thomas.** 2006. *Elements of Information Theory*. Hoboken, NJ: John Wiley & Sons.
- DellaVigna, Stefano, John A. List, and Ulrike Malmendier.** 2012. "Testing for Altruism and Social Pressure in Charitable Giving." *Quarterly Journal of Economics* 127 (1): 1–56.
- DellaVigna, Stefano, John A. List, Ulrike Malmendier, and Gautam Rao.** 2017. "Voting to Tell Others." *Review of Economic Studies* 84 (1): 143–81.
- Edwards, Allen L.** 1957. *The Social Desirability Variable in Personality Assessment and Research*. Hinsdale, IL: Dryden Press.
- Folsom, Ralph E., Bernard G. Greenberg, Daniel G. Horvitz, and James R. Abernathy.** 1973. "The Two Alternate Questions Randomized Response Model for Human Surveys." *Journal of the American Statistical Association* 68 (343): 525–30.
- Genicot, Garance.** 2022. "Tolerance and Compromise in Social Networks." *Journal of Political Economy* 130 (1): 94–120.
- Gerber, Alan S., Donald P. Green, and Christopher W. Larimer.** 2008. "Social Pressure and Voter Turnout: Evidence from a Large-Scale Field Experiment." *American Political Science Review* 102 (1): 33–48.
- Greenberg, Bernard G., Roy R. Kuebler, James R. Abernathy, and Daniel G. Horvitz.** 1971. "Application of the Randomized Response Technique in Obtaining Quantitative Data." *Journal of the American Statistical Association* 66 (334): 243–50.
- Greenberg, Sarah, and F. Daniel Stone.** 2024. "Self-Censorship in the Classroom." Unpublished.
- Inbar, Yoel, and Joris Lammers.** 2012. "Political Diversity in Social and Personality Psychology." *Perspectives on Psychological Science* 7 (5): 496–503.
- Kahneman, Daniel.** 2012. *Thinking, Fast and Slow*. New York, NY: Penguin.
- Karing, Anne.** Forthcoming. "Social Signaling and Childhood Immunization: A Field Experiment in Sierra Leone." *Quarterly Journal of Economics*.
- Kartik, Navin.** 2009. "Strategic Communication with Lying Costs." *Review of Economic Studies* 76 (4): 1359–95.
- Kosfeld, Michael, and Susanne Neckermann.** 2011. "Getting More Work for Nothing? Symbolic Awards and Worker Performance." *American Economic Journal: Microeconomics* 3 (3): 86–99.
- Krumpal, Ivar.** 2013. "Determinants of Social Desirability Bias in Sensitive Surveys: A Literature Review." *Quality and Quantity* 47: 2025–47.
- Lea, John.** 2009. *Political Correctness and Higher Education*. Abingdon, Oxfordshire: Routledge.
- Loury, Glenn C.** 1994. "Self-Censorship in Public Discourse." *Rationality and Society* 6 (4): 428–61.
- Morris, Stephen.** 2001. "Political Correctness." *Journal of Political Economy* 109 (2): 231–65.
- Niche.** 2020. Most Liberal Colleges in America. <https://www.niche.com/colleges/search/most-liberal-colleges/>.
- Perez-Truglia, Ricardo, and Guillermo Cruces.** 2017. "Partisan Interactions: Evidence from a Field Experiment in the United States." *Journal of Political Economy* 125 (4): 1208–43.
- Perez-Truglia, Ricardo, and Ugo Troiano.** 2018. "Shaming Tax Delinquents." *Journal of Public Economics* 167: 120–37.
- Raghavarao, D., and W. T. Federer.** 1979. "Block Total Response as an Alternative to the Randomized Response Method in Surveys." *Journal of the Royal Statistical Society* 41 (1): 40–45.
- Roth, Michael S.** 2019. *Safe Enough Spaces: A Pragmatist's Approach to Inclusion, Free Speech, and Political Correctness on College Campuses*. New Haven, CT: Yale University Press.

- Schwarz, Gonzalo.** 2020. Interview with James Heckman. *Medium*. <https://medium.com/archbridge-notes/nobel-prize-winning-economist-dr-5550da1df5c3>.
- Sniderman, Paul M., and Thomas Piazza.** 1993. *The Scar of Race*. Cambridge, MA: Harvard University Press.
- Sudman, Seymour, and Norman M. Bradburn.** 1974. *Response Effects in Surveys*. Chicago, IL: Aldine Publishing Company.
- Tourangeau, Roger, Lance J. Rips, and Kenneth Rasinski.** 2000. *The Psychology of Survey Response*. Cambridge: Cambridge University Press.
- Warner, Stanley.** 1965. "Randomized Response: A Survey Technique for Eliminating Evasive Answer Bias." *Journal of the American Statistical Association* 60 (309): 63–69.
- Warner, Stanely.** 1971. "The Linear Randomized Response Model." *Journal of the American Statistical Association* 66 (336): 884–88.
- Zimmer, Robert J.** 2016. "Free Speech Is the Basis of a True Education." *Wall Street Journal*, August 26. <https://www.wsj.com/articles/free-speech-is-the-basis-of-a-true-education-1472164801>.